

Synthetic Relativity: Tri-modality and the Fixpoint Semantics of the Logical Observer

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Introduction to Synthetic Relativity

Synthetic Relativity

Scope of talk

- Synthetic relativity as in synthetic geometry, or synthetic domain theory, i.e. *axiomatic*
- **Focus:** Self-reflection and reasoning about theories of action

Out of scope

This is *not* a first-order logic foundation of physical relativity theories with accelerated observers e.g. work by the *Budapest Relativity Research Group* (Madarász et al. 2006, Székely et al. 2010)

- André et al., 2013 provides an axiomatisation complete for Lorentzian manifolds, i.e. the Lorentz interval invariant $(c\Delta t)^2 - distance^2$ always equals zero.

Self-reflective reasoning about action

Motivation: Understand role of self-reflection for reasoning about action, for tasks such as

- Classical planning,
- Generalised planning,
- Strategic reasoning (strategy synthesis, winnability, etc.).

Reach goal:

- Self-transcending reasoning, i.e. **autonomous theory progression for transcending a theory of action toward solving *new problems***

Transformation example in logic

The space-time tradeoff **Memoisation** versus **lazy evaluation**.

Memoisation (Time $\rightarrow$ Space)

You take a proof that requires coinductive steps (Time), evaluate it in advance, and cache the valid answers into a lookup table (Space/Induction)—You have contracted logical time (proof) and dilated logical space.

Lazy Evaluation (Space $\rightarrow$ Time)

You take memory (Space/Induction) and collapse it. Instead of storing the data, you store a coalgebraic rule that generates proofs sequentially, only when requested (Time/Coinduction)—You have contracted logical space and dilated logical time (proof).

The logical observer

Different logical observers evaluate systems differently:

- **One observer** (evaluating lazily/**coinductively**) stretches out the “time” of the proof.
- **Another observer** (evaluating eagerly/**inductively**) compresses the proof into memory “space”.

However, the fundamental computational truth—**the realisation**—cannot change.

If truth were not invariant under these frame-shifts, logic would collapse into inconsistency. The spatial distortion of the data structure and the temporal delay in evaluation mathematically cancel out to preserve the invariant **realisation**.

Logical interval invariant?

In synthetic relativity, inductive structure and coinductive process are two mathematically intertwined ways of evaluating the same invariant truth.

- **Induction** (static space/structure) and
- **Coinduction** (unfolding time/process) are the coordinate axes, and
- **Realisation** is the invariant interval.

Rotating logical frame of reference changes how much proof exists in spatial topology versus how much unfolds dynamically in computational time, but invariant interval—the **realisation**—is preserved.

- Note in the **situation calculus** that since *regression* is trivially **confluent**, as per Church-Rosser theorem as it is deterministic, the order in which you evaluate it does not change its final semantic truth, so the **realisation** is preserved.

Inference and Categorical Duality

Inference duality of the observer

Logical Benders decomposition (Hooker & Ottosson, 2003) shows how the observer iteratively proposes a model of itself and verifies it via self-reflection.

- **The Master (The Proposer):** The observer proposes a relaxed model of its current state or plan. **Operator (action) counting using MIP**
- **The Subproblem (The Verifier):** The observer reflects on this proposal, testing its logical consistency with respect to the overall problem. **Sequencing actions using SAT**
- **The Cut (The Inference):** When the proposal fails, the observer generates a logical inference (a Benders cut) that restricts its future reasoning. **Conflict driven clause learning (SAT).**

(Davies, Pearce, Stuckey & Lipovetzky 2015, ICAPS Outstanding Paper Award) showed approach competitive with state of the art planners on International Planning Competition (IPC) benchmarks—proven to converge to the optimal feasible solution.

Inference duality of the observer - Limitations

Inference duality like Benders decomposition can be modelled as finding a fixpoint of a monotonic constraint-generation operator.

- However, it does not compute a “least fixpoint” in the traditional sense as it intentionally stops early once optimal feasibility is proven (it is based on propositional logic)
- Note that the **situation calculus** *does* rely on least fixpoint in first order logic.

Property Persistence as a Greatest Fixpoint

The Challenge: Verifying universally quantified temporal queries (e.g., “ ϕ persists indefinitely”) requires second-order induction, which strictly limits automated first-order reasoning.

The Single-Step Operator: (Kelly & Pearce, AIJ 2010) captures the condition for ϕ holding *now* and being preserved after *any single legal action* via regression ($\mathcal{R}$):

$$\Gamma(Z) \equiv \phi(s) \wedge \forall a (Poss(a, s) \rightarrow \mathcal{R}[Z(do(a, s))])$$

Greatest Fixpoint (νZ):

- Persistence is an infinite-horizon **safety property**.
- The persistence condition is the **greatest fixpoint** of this operator:
 $\nu Z. \Gamma(Z)$.

Iteratively applying this meta-level calculation computes the maximal invariant without explicitly invoking the induction axiom.

- By wrapping the single-step regression operator inside a monotonic formula transformer, we evaluate the future back in the current state.
- Taking the greatest fixpoint of this regression operator yields the exact, most permissive preconditions required for the property to persist indefinitely using only first-order logic.

Strategy Synthesis in Multi-Agent Games

- **Scaling Fixpoint Framework:** Property persistence scales natively from single-agent persistence to adversarial settings via *Situation Calculus Synchronous Game Structures (SCSGSs)*.
- **General Game Playing (GDL):** (De Giacomo, Lespérance & Pearce, ECAI 2016) model joint, synchronous multi-player moves over infinite domains via a single `tick(m_1, ..., m_n)` action.
- **First-Order Alternating-Time μ -Calculus:**
 - **Safety:** Greatest fixpoints (νZ) — Generalising persistence to maintain a winning condition against any opponent (*Weak Winnability*).
 - **Liveness:** Least fixpoints (μZ) — Enforcing reachability to force a win in a finite number of steps (*Strong Winnability*).
- **Synthesis:** By upgrading the quantifier prefix to alternating game quantifiers ($\exists \vec{m}_{agt} \forall \vec{m}_{env}$), interleaving regression with fixpoint approximations automates strategy synthesis.

The modal μ -calculus derives its expressive power from the same foundational fixpoint mathematics—least and greatest fixpoints—used to compute backward reachability and property persistence in model checking.

- By upgrading the quantifier prefix from ‘nature does anything’ to **alternating game quantifiers**, we can use both greatest and least fixpoints to automatically synthesise strategies and guarantee winnability in infinite-state games.

Categorical duality

Is property persistence just an algorithmic trick with regression? No. It represents a fundamental categorical duality in action theories.

- **The Inductive Action Theory (Standard SitCalc):**
 - Builds state histories forward from S_0 .
 - Categorically maps to **Initial Algebras** and **Least Fixpoints** (μ).
 - Naturally solves *reachability* and *liveness*.
- **The Coinductive Persistence Theory (AIJ 2010):**
 - Persistence is an infinite-horizon safety property.
 - Categorically maps to **Terminal Coalgebras** and **Greatest Fixpoints** (ν).
 - Employs **Regression** (the *right adjoint* to progression) to compute maximal invariants.

By framing persistence as a greatest fixpoint over single-step regression, we embedded the **coinductive dual** of reachability into the Situation Calculus.

- This mathematical duality made the transition to alternating μ -calculus for game synthesis (ECAI 2016) a natural evolution.

In retrospect, what we were doing in 2010 was exploiting a **categorical duality**. The standard Situation Calculus is fundamentally inductive—an initial algebra. But persistence is fundamentally coinductive—a terminal coalgebra. By evaluating persistence via regression, we mapped a forward-looking infinite problem into its computable, right-adjoint dual.

- Once you formalise the dual spaces of safety (greatest fixpoints over regression) and reachability (least fixpoints), strategy synthesis in the full μ -calculus is just the natural interleaving of these mathematical duals.

Limitations

What was the limitations of our approach?

- Regression is handled *separately* by different logics: μ calculus (greatest fixpoint μ and least fixpoint ν) + situation calculus (least fixpoint μ)
- Neither the μ calculus nor the situation calculus captures *inference duality* or *categorical duality* explicitly
- They rely on a hand-coded basic action theory (BAT)

Recall **Stretch goal**: self-transcending reasoning

- A relativistic framework for reasoning about limits and topological structure
- A framework for the autonomous progression of action theories
- A partial order approach to both synthesising and hypothesising the existence of limits

Synthetic relativity axioms

Tri-modality & ternary representation

Standard provability logics like GLB (Gödel-Löb Bimodal Provability Logic) rely on absolute binary provability: xRy assumes an absolute background space where y is provable from x .

- In a ternary structure (drawing from Routley-Meyer semantics for substructural, relevance, and intuitionistic logics), you have a relation $R(x, y, z)$.
- This allows us to decouple the synthetic transformation from the topological space. The ternary relation natively introduces the Observer's frame. It reads: "Relative to reference frame x , the accumulation of inductive evidence y yields the coinductively realised semantic limit z ."

As the semantic space is no longer strict and binary, the representational geometry must evolve to match the problem.

- **The Tri-modal Shift:** Introduces a ternary representation to dynamically navigate the relativistic observer.
- **Structural Necessity:** Tri-modality is the specific geometry required to house the non-standard continuous spaces.
- **The Outcome:** The observer can achieve self reflection (coinduction) while formally stepping outside its previous logical bounds (induction).

Synthetic Relativity: The Structural Axioms

Dynamic Trimodal Logic with Ternary Relational Semantics:

Axiom 1: The Ternary Relational Axiom Logical entailment is governed by three interacting modalities relative to a specific baseline Reference Frame (S_f):

- **RDo (The Subproblem / Syntax): The Inductive Accumulator ($\Box_1$)** Operates in constructive, non-Boolean logic. Recursively gathers finite, computationally observable properties (compact elements) along a directed logical path. RDo is equivalent to the $\Box$ modality of bimodal provability logic GLB: “*There exists a finite, step-by-step mechanical proof of this property.*”
- **ODo (The Master Problem / Semantics): The Limit/Creator Modality ($\Box_2$)** The semantic evaluator and *interval invariant*. Performs topological closure—evaluating the directed set generated by RDo and snapping it into a discrete, continuous limit. ODo is equivalent to the $\Box_\omega$ modality of GLB: “*Evaluated at the continuous limit, this property is universally true,*” i.e., this is the modality of *semantic reflection*.
- **PDo (Autonomous Progression): The Relativistic Engine ($\Box_3$)** The dynamic transition operator. It orchestrates the recursive interaction of local syntax (RDo) and global semantics (ODo) relative to S_f .

Constructive Accumulation & Realisation

Axiom 2: The Axiom of Constructive Accumulation Within a given reference frame (S_f), the Relativistic Engine (PDo) iteratively drives the state forward, nesting inductive syntax (RDo) and semantic coinductions (ODo):

$$\text{PDo}(S_r, S_o, S_f)$$

$$\text{PDo}(\text{RDo}(R_1, S_r), \text{ODo}(O_1, S_o), S_f)$$

$$\text{PDo}(\text{RDo}(R_2, \text{RDo}(R_1, S_r)), \text{ODo}(O_2, \text{ODo}(O_1, S_o)), S_f)$$

etc.

Theorem I: Semantic Realisation

Because progression builds strictly upon finite verifiable observables, autonomous progression strictly halts when the topology reaches Algebraic Compactness. This is the moment of mathematical **Realisation**:

$$\mu(\text{PDo}) = \text{ODo} \left(\bigsqcup_{i \in R} \text{RDo}_i(\perp) \right)$$

- $\bigsqcup \text{RDo}$ (**Syntax**): The builder generates a raw, directed chain of evidence from the ground up. A syntax string alone has no inherent “meaning.”
- ODo (**Semantics**): The top-down evaluator reads the syntax trace and evaluates its truth value.

Proof Sketch: Corresponds to the computation of the **Kleene Fixed-Point Theorem**.

- The ultimate stable fixpoint (μ) is caught exactly when ODo evaluates the limit (Supremum, $\bigsqcup$) of iteratively applying the recursively enumerable inductive proofs (RDo) from baseline ignorance ($\perp$).
- This realised limit acts as the interval semantic invariant.

The Synthetic Transformation

Axiom 3: The Transitivity Axiom Once Realisation is achieved, the relativistic reference frame shifts. The categorically preserved invariant (the semantic limit, O) becomes the new reference frame (S_f) for the next order of progression.

$$\text{PDo}(N, O, S_f) \longrightarrow \text{PDo}(X, Y, \mathbf{O})$$

- **Left Side:** The fully realised state within the initial reference frame (S_f).
- **The Arrow ($\longrightarrow$):** The Synthetic Transformation (Topological closure and Change of Base).
- **Right Side:** A new, higher-order autonomous progression begins. The complex limit O has mathematically synthesised into an atomic, discrete baseline coordinate for the new subproblem (X).

Theorem II: Relativistic Invariance (Preserving the Interval)

Proof Sketch: Category theory mathematically guarantees that under this Transitivity shift (a geometric morphism), the overarching semantic truth-value and geometric integrity of the realised limit (O) are strictly and losslessly preserved across the Change of Base.

The Dual Fixpoints

Unifying Reachability (μ) and Persistence (ν)

By natively intertwining syntax (RDo) and semantics (ODo), the Relativistic Engine mathematically fuses two distinct forms of logical progression:

1. **Induction / Least Fixpoint (μ):** Governed by the RDo modality. The bottom-up synthesis of finite evidence from baseline ignorance ($\perp$). It guarantees computational *Reachability*.
2. **Coinduction / Greatest Fixpoint (ν):** Governed by the ODo modality. The top-down evaluation of continuous semantics. It guarantees *Safety and Property Persistence*.

Algebraic Compactness ($\mu = \nu$) Theorem I (Semantic Realisation) formally computes a **Bilimit**. At the exact coordinate of convergence, the active inductive trace (μ) perfectly fulfills the persistent semantic invariant (ν).

*When this limit shifts to become the new Reference Frame (S_f) via the Transitivity Axiom, **Theorem II** guarantees it acts entirely as a Greatest Fixpoint—a persistent, assumed law of physics for all subsequent computations, natively solving the Frame Problem.*

Synthetic relativity logic

The Observer's Dual Logic (Syntax vs. Semantics)

To execute our modalities, we require an architecture that natively fuses classical (inductive) logic with intuitionistic (coinductive) logic.

- **Memoisation (Induction / The Actor):** Constructive in the *data* sense. It explicitly builds and stores finite, historical knowledge. It captures logical invariants and acts as the proof of what has been physically done.
- **Lazy Evaluation (Coinduction / The Observer):** Constructive in the *intuitionistic* sense. The runtime intuitively maintains the topology of an infinite stream, only verifying when locally forced.

The Incompleteness Boundary: Logic remains classically monotonic locally. However, due to relative incompleteness (Gödel's Incompleteness & Beklemishev's reflection principles), a static system cannot resolve global limits. The observer must step outside the local system.

The Foundational Logic: Domain Theory in Logical Form

To map this topology of reasoning, we ground our system in Abramsky's **DTLF** (1991), based on an **intuitionistic** construction of truth.

- **Baseline Ignorance ($\perp$):** Computation initiates at Bottom ($\perp$)—the mathematical coordinate of zero information.
- **Compact Elements (Finite Observability):** Basic propositions serve as atomic, observable quanta. In DTLF, logical formulas correspond directly to compact elements, meaning any verified property must be confirmable in finite time.
- **Directed Accumulation ($\sqsubseteq$):** Knowledge grows monotonically. The logic is strictly positive and constructive, meaning once an observation is verified, it establishes an upward, directed path toward the limit without retraction.

Spatial Completeness & The Static Limitation

The mathematical power of vanilla DTLF lies in **Spatial Completeness** (rooted in Stone Duality).

- **The Guarantee:** It proves a 1-to-1 isomorphism between the algebraic, step-by-step **Syntax** ($\phi \vdash \psi$) and the continuous, geometric **Semantics** ($\llbracket \phi \rrbracket \subseteq \llbracket \psi \rrbracket$).
- **Catching Limits:** Computations resolve exactly when the directed path of finite evidence mathematically catches (satisfies) an infinite continuous topological limit (a Supremum).

Vanilla DTLF perfectly describes a single, *static* universe. However, to differentiate induction from coinduction and model dynamic, continuous learning, **the Reference Frame itself must dynamically shift.**

Encoding Planning in Synthetic Relativity (R-DTLF)

Classical planning (e.g., Situation Calculus) relies on a global, static Basic Action Theory (BAT), where rigid evaluation across absolute history leads to combinatorial explosion. In **R-DTLF**, planning is constructed dynamically:

- **Initial Database** $\rightarrow$ **Reference Frame** (S_f): Dictates the local inductive bias—bounding exactly which *fluents* (e.g., **Disk positions in Towers of Hanoi**) are currently observable.
- **The Situation Term** ($do(a, s)$) $\rightarrow$ **Inductive Accumulation** (RDo): Instead of an endlessly nested string of rigid state-jumps, actions are encoded as verifiable proof steps recursively accumulating along a directed path.
- **Goal Formula** ($\exists s. G(s)$) $\rightarrow$ **Semantic Target** (ODo): The goal is not a static Boolean theorem. It is encoded as a **Target Semantic Attractor**—the fluent configuration or overarching invariant the system coinductively attempts to catch.
- **Deductive Search** $\rightarrow$ **Relativistic Engine** (PDo): The autonomous, recursive unrolling of these modalities.

Execution & Invariant Synthesis: Towers of Hanoi example

1. **Local Unrolling (Building the Situation Trace):** The engine recursively accumulates primitive actions (**moving disks**), generating the topological equivalent of a nested $do()$ sequence:

$$\text{PDo}\left(\text{RDo}(\text{move}(D_1, P_2), \dots S_f), \text{ODo}(O_{\text{target}}, \text{Goal}), S_f\right)$$

2. **Limit Catching:** Through top-down semantic reflection ($\square_\omega$), the ODo modality mathematically coinduces a structural property of the optimal traces: **The Parity Invariant (O_{parity})**. *(The smallest disk D_1 must cycle consistently in one fixed direction, while every alternate move is the single legal play between the remaining two pegs, never involving D_1).*

3. **The Synthetic Transformation (Change of Base):** The Transitivity Axiom fires, executing a categorical Change of Base:

$$\text{PDo}(N_{\text{trace}}, O_{\text{parity}}, S_f) \longrightarrow \text{PDo}(X, Y, \mathbf{O}_{\text{parity}})$$

Why is the parity invariant found in synthetic relativity (R-DTLF)?

- The alternate move is discovered as the only topologically valid way to accumulate new information.
- In topological terms, reversing the cycle leads to a “dead end” in the domain—a state that cannot act as a supremum for a directed set leading to the goal limit.
- The invariant (cycle D_1 , alternate move, cycle D_1) emerges as the unique, strictly positive proof path through the state space.

In standard classical planners, the Towers of Hanoi puzzle has a continuous branching factor of 3, leading to intractable combinatorial explosion for deep goals.

- The new parity invariant shifts to become the Reference Frame. By adopting this rule as a **lifted axiom**, *the search tree branching factor strictly collapses from 3 down to exactly 1.*
- Note computational complexity of problem doesn't, just *where* the computational effort is spent between a classical recursive approach versus Synthetic Relativity (DTLF) progression.

Translating Continuous Syntax into Successor State Axioms

1. Sidestepping the Frame Problem (SSA vs. R-DTLF)

- **SitCalc SSA (Global Tracking):** Evaluates the entire history (s) across rigid Successor State Axioms:

$$On(d, p, do(a, s)) \equiv a = \text{move}(d, p) \vee (On(d, p, s) \wedge \neg \exists p'. a = \text{move}(d, p'))$$

- **R-DTLF (Local Entailment):** Operates purely on verifiable observables. Irrelevant global variables are topologically invisible (no frame axioms are computed during unrolling):

$$RDo(\text{move}(d, p), S_f) \vdash On(d, p)$$

2. Endogenous Code Generation (Compiling the Precondition) Once O_{parity} is realised, the engine autonomously exports the invariant into a discrete classical **Action Precondition Axiom** ($Poss$) for the new baseline:

$$Poss_{\text{new}}(\text{move}(d, p), s) \equiv Poss_{\text{old}} \wedge (\mathbf{d} = \mathbf{D}_1 \leftrightarrow \text{LastMoved}(s) \neq \mathbf{D}_1)$$

By autonomously synthesising this SSA, the logic enforces deterministic execution.

Bridging Worlds: Endogenous Logic Compilation

R-DTLF does not discard classical logic; it functionally *generates* it at the semantic limits.

Theorem III: Topological Booleanization (Compiling Successor State Axioms)

Upon semantic realisation, the continuous intuitionistic trace generated by $\mathbf{RDo}$ can be losslessly compiled into a discrete, classical First-Order baseline via the **Lawvere-Tierney Double Negation ($\neg\neg$) Topology**.

Proof Sketch: Topos theory formally guarantees that inside every fluid, intuitionistic space exists a rigid Boolean sub-lattice that can be safely projected onto.

- **Compiling Successor State Axioms (SSAs):** This geometric compression is mathematically homologous to Ray Reiter's **Explanation Closure**. The Relativistic Engine acts as an endogenous logic compiler—synthesising complex discovered invariants directly into rigid First-Order logical rules.
- **Hybrid Execution:** Because of this formal theorem, we can securely export newly learned R-DTLF heuristics directly into standard classical Basic Action Theories (BATs). This allows R-DTLF to handle complex structural learning, and legacy solvers (like Golog).

Comparison

Synthetic Relativity versus Situation Calculus versus GLB

Aspect	<i>Synthetic Relativity (R-DTLF)</i>	<i>Situation Calculus (BATs/SSAs)</i> + <i>variants of μ-calculus</i>	<i>Bimodal Provability logic (GLB)</i>
Modalities	RDo, ODo & PDo	PDo, $\langle a \rangle \phi$ (possibility), $[a] \phi$ (necessity)	$\Box, \Box_\omega$
Ordering	Sequences of finite Posets, non-unique predecessors, non-well ordered	Successor function, Unique predecessors, non-well ordered	Transfinite ordinality: limits, Non-unique predecessors, well-ordered
Completeness?	Spatially complete (continuous geometry) and Turing complete (universal computation via recursive domains); Arithmetically incomplete (triggers Gödel's incompleteness).	Turing complete (can compute anything), but neither spatially complete (no continuous geometry) nor arithmetically complete (subject to Gödel's incompleteness).	Arithmetically complete (flawlessly models the bounds of provability), but neither Turing complete (strictly decidable, cannot compute loops) nor spatially complete.
Other	Algebraic compactness. Kleene's Realizability & BHK (Brouwer-Heyting-Kolmogorov) Interpretation.	Regression Theorem (solves the frame problem). Fixpoint Semantics & Knaster-Tarski Theorem.	Japaridze's Arithmetical Completeness Theorem. Löb's Theorem & Well-founded Kripke Semantics.